

Wireshark Network Protocol Analysis

HTTP Traffic Analysis

A pre-captured HTTP packet capture was analysed using Wireshark to investigate HTTP communication. Packets 25–33 were examined to identify the application and transport layer protocols, analyse addressing information, and observe the exchange of HTTP requests and responses.

Protocol Identification

The captured traffic uses HTTP as the application layer protocol. HTTP provides a request-response service that enables the transfer of web pages and other resources between clients and web servers.

The transport layer protocol used is TCP which provides reliable, connection-oriented communication between devices.

HTTP Request Packet (Frame 27)

Source IP Address	51.128.215.10
Destination IP Address	51.128.224.11
HTTP Version	HTTP/1.0
Source Port	446168
Destination Port	80

HTTP Response Packet (Frame 29)

Source IP Address	51.128.224.11
Destination IP Address	51.128.215.10
HTTP Version	HTTP/1.0
Source Port	80
Destination Port	446168

Analysis and Observations

The packets show normal HTTP communication between a client and a server.

In frame 27, the client (51.128.215.10) sends a request to the server (51.128.224.11) using HTTP version 1.0. The server uses the standard port of 80 for HTTP, while the client uses a temporary port (446168).

In frame 29, the server responds back to the client. The source and destination IP addresses and port numbers are reversed, indicating a reply to the same session. This demonstrates a request-response process and confirms successful communication between the client and server.

SMTP Traffic Analysis

A pre-captured SMTP packet capture was analysed using Wireshark to examine SMTP (Simple Mail Transfer Protocol) communication. Packets 12-35 were inspected to identify the protocols used, analyse packet information and observe how email communication is established between a client and a mail server.

Protocol Identification

The captured traffic uses SMTP as the application layer protocol. SMTP is responsible for sending email messages from a client to a mail server or between mail servers.

The transport protocol used is TCP (Transmission Control Protocol), which provides reliable, connection-oriented communication to ensure email data is delivered accurately.

SMTP Packet Analysis (Frame 26)

Source IP Address	10.1.1.31
Destination IP Address	10.1.1.72
Source Port	48604
Destination Port	25

Analysis and Observations

The captured packet demonstrates normal SMTP communication between a client and a mail server. The client (10.1.1.31) establishes a connection to the mail server (10.1.1.72) using the standard SMTP port (25), while the client communicates using an ephemeral source port (48604). The use of TCP provides reliable delivery and ensures that email data is transmitted successfully. Overall, the packet capture indicates that communication between the client and the mail server was successfully established.

DNS Traffic Analysis

A pre-captured DNS packet capture was analysed using Wireshark to examine DNS (Domain Name System) communication. Packets 12-14 were inspected to observe how domain names are resolved into IP addresses and to identify the protocols involved in the exchange.

Protocol Identification

The captured traffic uses DNS as the application layer protocol. DNS provides a name resolution service by translating domain names into IP addresses, allowing users to access websites using human readable names.

The transport layer protocol used is UDP (User Datagram Protocol), which provides fast, connectionless communication suitable for DNS queries.

DNS Query Packet (Frame 13)

Source IP Address	51.128.215.10
Destination IP Address	51.128.224.10
Source Port	44565
Destination Port	53

DNS Query Packet (Frame 14)

Source IP Address	51.128.224.10
Destination IP Address	51.128.215.10
Source Port	53
Destination Port	44565

Analysis and Observations

The packet capture demonstrates a standard DNS query-response exchange between a client and a DNS server. The client (51.128.215.10) sends a DNS query to the server (51.128.224.10) using an ephemeral source port (44565), while the server listens on the standard DNS port (53). The response packet reverses the source, and destination addresses and port numbers, confirming that the DNS server successfully processed the query and returned the requested information. This illustrates how DNS operates over UDP to provide fast and efficient domain name resolution.

HTTPS Traffic Analysis (my.gov.au)

During the Wireshark capture of <https://my.gov.au/>, network traffic was analysed to observe how the device communicates with external servers.

```
C:\Users\enagp>nslookup my.gov.au
Server:  NF18ACV.Home
Address:  192.168.20.1

Non-authoritative answer:
Name:     my.gov.au
Addresses: 2600:9000:20ec:6200:a:ae1c:5880:93a1
           2600:9000:20ec:4400:a:ae1c:5880:93a1
           2600:9000:20ec:b000:a:ae1c:5880:93a1
           2600:9000:20ec:3600:a:ae1c:5880:93a1
           2600:9000:20ec:e600:a:ae1c:5880:93a1
           2600:9000:20ec:a400:a:ae1c:5880:93a1
           2600:9000:20ec:600:a:ae1c:5880:93a1
           2600:9000:20ec:f800:a:ae1c:5880:93a1
           13.224.181.111
           13.224.181.99
           13.224.181.109
           13.224.181.97
```

The nslookup my.gov.au command returned multiple IP addresses, showing that the website is hosted on multiple servers. This suggests a distributed infrastructure, used for load balancing and reliability so that user requests can be handled efficiently across different servers.

```

C:\Users\enagg>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 4:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::e816:d66f:d57:ee6d%20
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Wireless LAN adapter Local Area Connection* 9:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : Home
    Link-local IPv6 Address . . . . . : fe80::377a:ab9c:162a:1cfc%23
    IPv4 Address. . . . . : 192.168.20.12
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::faca:59ff:fe32:c09%23
                                192.168.20.1

```

The ipconfig command showed the local device IP address as 192.168.20.12, which identifies my computer on the local network. This IP is only valid internally and is used for communication within the local network before traffic is routed externally.

Analysis and Observations

The HTTPS capture demonstrates how secure web communication differs from HTTP. While HTTP traffic can be inspected directly, HTTPS encrypts the application-layer data, protecting the contents of requests and responses. Only addressing information, protocol headers and transport details remain visible during packet analysis.